

# Credit-Card Fraud Detection — Summary Report

**Dataset:** ULB creditcard.csv — 284,807 transactions over 2 days; after removing 1,081 exact duplicates, **283,726 rows, 473 fraud (0.167%, ≈600:1 imbalance)**. Features: Time (elapsed seconds), V1–V28 (anonymized **PCA** components), Amount, Class (target).

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## 1. Executive summary

Two gradient-boosted tree models — **XGBoost** and **LightGBM** — both reach **AUPRC ≈ 0.84** on leakage-free 5-fold cross-validation, against a no-skill baseline of 0.0017. A wider search (CatBoost, stacking, anomaly detection, web-researched “SOTA” hybrids/transformers) did **not** materially beat them: for a 280k-row *anonymized-PCA* table, tuned gradient boosting is the frontier. The two models are deliberately complementary — **XGBoost leans recall, LightGBM leans precision** — which is the main reason we keep both.

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## 2. Methodology

The extreme imbalance dictates every decision; the pipeline is built to be **leakage-free** end to end.

1. **Deduplicate first.** 1,081 exact duplicates are dropped *before* any split — duplicates that straddle a train/test boundary leak minority information and inflate density.
2. **Feature engineering (minimal, deliberate).**
  - V1–V28: **left untouched** — already PCA (orthogonal, centered, no domain meaning); no scaling, no interactions, no pruning.
  - Amount: kept raw for trees (scale-invariant); log1p + RobustScaler path available for non-tree models.
  - Time → **cyclic hour-of-day** (sin/cos); raw Time dropped (it is a 48-hour capture ramp, not a timestamp, and leaks collection order).
3. **Validation.** Stratified 5-fold CV with **out-of-fold** predictions; scale\_pos\_weight computed **per fold** from training labels only. Imbalance handled by **class weighting, not SMOTE** (synthetic points on orthogonal PCA dims distort calibration).
4. **Metrics.** **AUPRC (Average Precision)** is primary; ROC-AUC reported but flagged as optimistic. The decision threshold is tuned on the **PR curve by F2** (recall-weighted), never the default 0.5.
5. **Deployment realism.** A **day-1 / day-2 temporal holdout** (train earlier, test later, threshold fixed from train) supplements random CV.

*Process note:* model choice, feature claims, and the leakage audit were each cross-checked by an independent multi-agent review and reconciled before being accepted.

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## 3. Why two models (and why these two)

- **Right family.** Gradient-boosted trees dominate numeric tabular data with a tiny positive class: scale-invariant, strong rare-class ranking, a single imbalance knob, fast on a GPU.
- **Diversity buys confidence and flexibility.** XGBoost (level-wise growth) and LightGBM (leaf-wise growth) are two *independent* implementations. When they agree, confidence rises; because they fail differently, they can be blended, and they expose **two operating profiles** (below).

- **Verified ceiling.** We empirically confirmed nothing better: CatBoost tied (0.841), an XGB+LGBM+CatBoost stack added only +0.001 (noise), Isolation Forest was far behind (0.14), and TabPFN/graph/VAE “SOTA” claims don’t apply here (row count out of regime; no relational structure in anonymized PCA). So two well-understood GBDTs is the correct scope — not five.

## 4. Model comparison

**5-fold out-of-fold (threshold tuned by F2 on the PR curve):**

| Metric                 | XGBoost       | LightGBM             |
|------------------------|---------------|----------------------|
| <b>AUPRC</b> (primary) | 0.842 ± 0.014 | <b>0.845 ± 0.015</b> |
| ROC-AUC                | <b>0.983</b>  | 0.981                |
| <b>Recall</b>          | <b>0.835</b>  | 0.827                |
| <b>Precision</b>       | 0.861         | <b>0.891</b>         |
| F2 (recall-weighted)   | <b>0.840</b>  | 0.839                |
| MCC                    | 0.847         | <b>0.858</b>         |
| TP / FP / FN           | 395 / 64 / 78 | 391 / <b>48</b> / 82 |

**Temporal day-1/day-2 holdout (deployment-realistic):**

| Metric       | XGBoost              | LightGBM      |
|--------------|----------------------|---------------|
| AUPRC        | 0.777                | <b>0.781</b>  |
| Precision    | <b>0.924</b>         | 0.911         |
| Recall       | <b>0.635</b>         | 0.630         |
| TP / FP / FN | 134 / <b>11</b> / 77 | 133 / 13 / 78 |

**Reading it:** ranking quality (AUPRC) is a statistical tie. The models separate at the *operating point*: **XGBoost catches more fraud (4 extra TP) at the cost of more false alarms (16 extra FP); LightGBM raises fewer false alarms but misses a few more frauds.**

## 5. Why XGBoost is stronger at recall and LightGBM at precision

The split comes from how each model is configured for the imbalance, plus how each grows trees:

- **XGBoost — recall-leaning.** It is trained with `scale_pos_weight` ≈ 599, i.e. each fraud is weighted ~600× a legit transaction. That pushes the decision boundary aggressively toward catching the minority class → **higher recall, more false positives**. Level-wise depth-5 trees give broad, high-coverage rules that flag more candidates.
- **LightGBM — precision-leaning.** It uses **no extra class weight** (`scale_pos_weight = 1`) — the naive ≈600 over-corrects its leaf-wise trees and was unstable (AUPRC collapsed to 0.59). With balanced weighting plus strong leaf regularization (`min_child_samples = 80`, `num_leaves = 31`), it only commits to fraud where the evidence is dense → **higher precision, fewer false positives**, at the cost of a few missed frauds.

**Intuition:** XGBoost is the *eager* detector tuned to “don’t miss fraud”; LightGBM is the *conservative* detector tuned to “don’t cry wolf.” This is partly a consequence of the imbalance-knob and F2-threshold choices (either model can be re-tuned to shift), but as configured the two cover the two business priorities — which is exactly the value of shipping both.

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## 6. Key findings (EDA)

- **Imbalance dominates.** All-legit accuracy is 99.83% and catches zero fraud → accuracy and ROC-AUC are misleading; AUPRC is the only honest headline.
  - **“Fraud = bigger amount” is false.** Fraud’s higher *mean* Amount (123.9 vs 88.4) is a tail artifact — its *median* is **lower** (9.82 vs 22.00).
  - **Fraud has a nightly rhythm.** By *rate*, fraud peaks around 02:00 (~145 per 10k) vs ~5 per 10k in business hours, while legit volume follows a day/night cycle — motivating the cyclic time feature.
  - **Signal is concentrated in a few PCA axes.** V14 alone is ~35–40% of XGBoost gain; the top-5 (V14, V10, V4, V17, V12)  $\approx$  64%. Keeping only the top-12 V’s matches all 28 within noise (0.832 vs 0.844), and Amount+time alone is near-zero (0.07) — the V’s carry essentially all the signal. (*Keep all features in production: univariate importance misses interactions.*)
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## 7. Recommendation

- **Default model:** either GBDT — pick by business cost. Use **XGBoost** when missing fraud is the expensive error (favor recall); use **LightGBM** when analyst review capacity / false-alarm cost is the constraint (favor precision). A simple probability blend captures both.
  - **Operate off the PR curve**, not 0.5: choose the threshold by the fraud-vs-false-alarm cost ratio (F2 is a reasonable default). Expect roughly the **temporal-holdout** numbers in production (~0.78 AUPRC, ~0.92 precision, ~0.63 recall), which are more realistic than random CV.
  - **Before final sign-off:** repeated (e.g. 3×5) stratified CV and a nested threshold would tighten the confidence intervals; the single-seed deltas here are within fold noise.
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*Artifacts:* fraud\_detection.ipynb (full code + analysis + plots, Colab-ready). Source scripts in the project root: fraud\_common.py, task1\_eda.py, fraud\_final.py, feature\_signal.py.